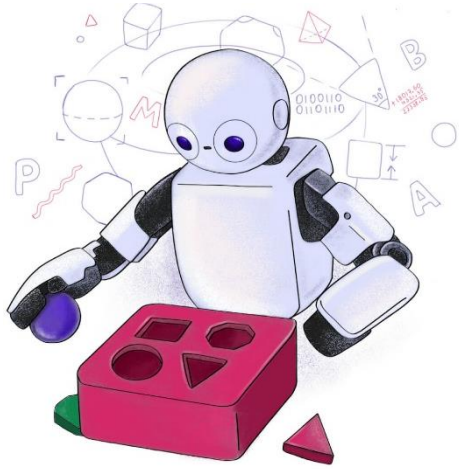
TP558 - Tópicos avançados em Machine Learning

Soft Actor-Critic

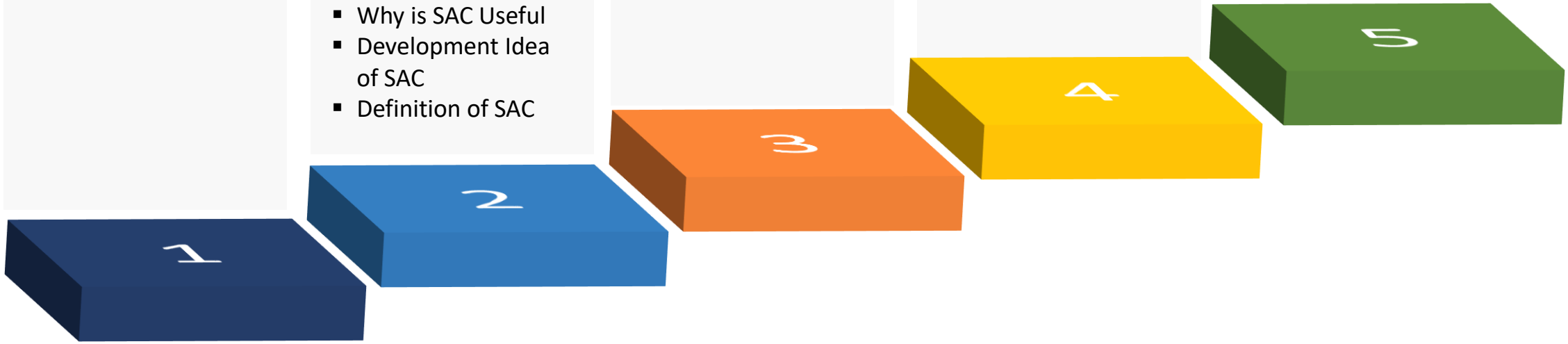


Inatel

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Outline

Introduction	Definition & Development of GNN	Challenges	Article	Future work & Conclusion
▪ Background ▪ Basic problem	▪ What is Soft Actor Critic ▪ Why is SAC Useful ▪ Development Idea of SAC ▪ Definition of SAC	▪ Deep RL side ▪ RIS-UAV side	▪ Algorithm ▪ Simulation Results	▪ Conclusion ▪ Future Directions

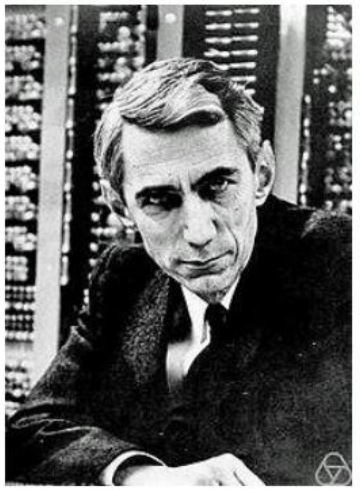


Introduction

What's New in 6G?

❑ **Three challenges for 5G operators**

- Number of 5G Bss
- Power consumption
- Hardwar cost



Claude Shannon（1916-2001）
"the father of information theory"

❑ Past decades: wireless channel H is only treated random and uncontrollable!

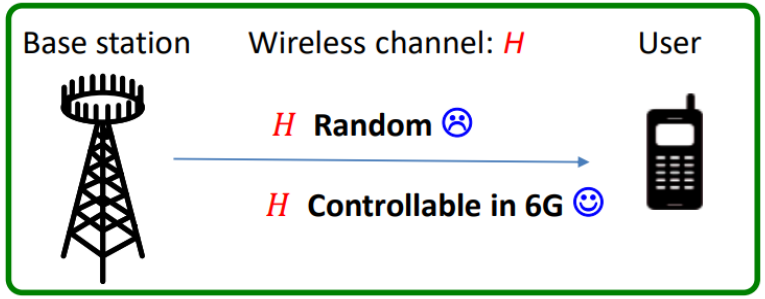
❑ Promising 6G paradigm: Smart and Reconfigurable Wireless Environment

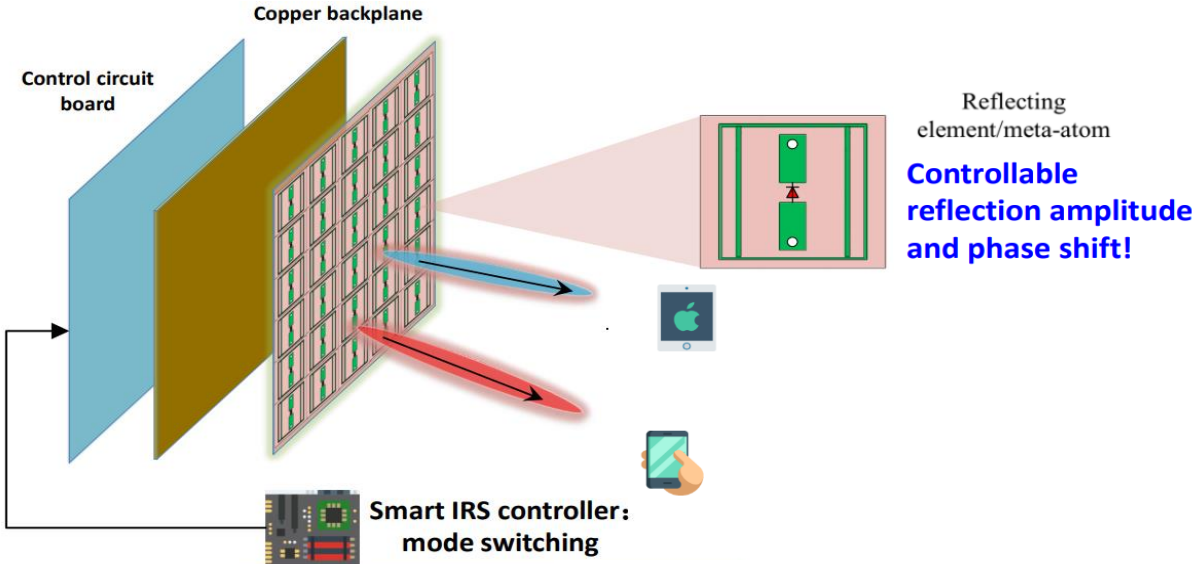
➢ New: Highly controllable wireless channel

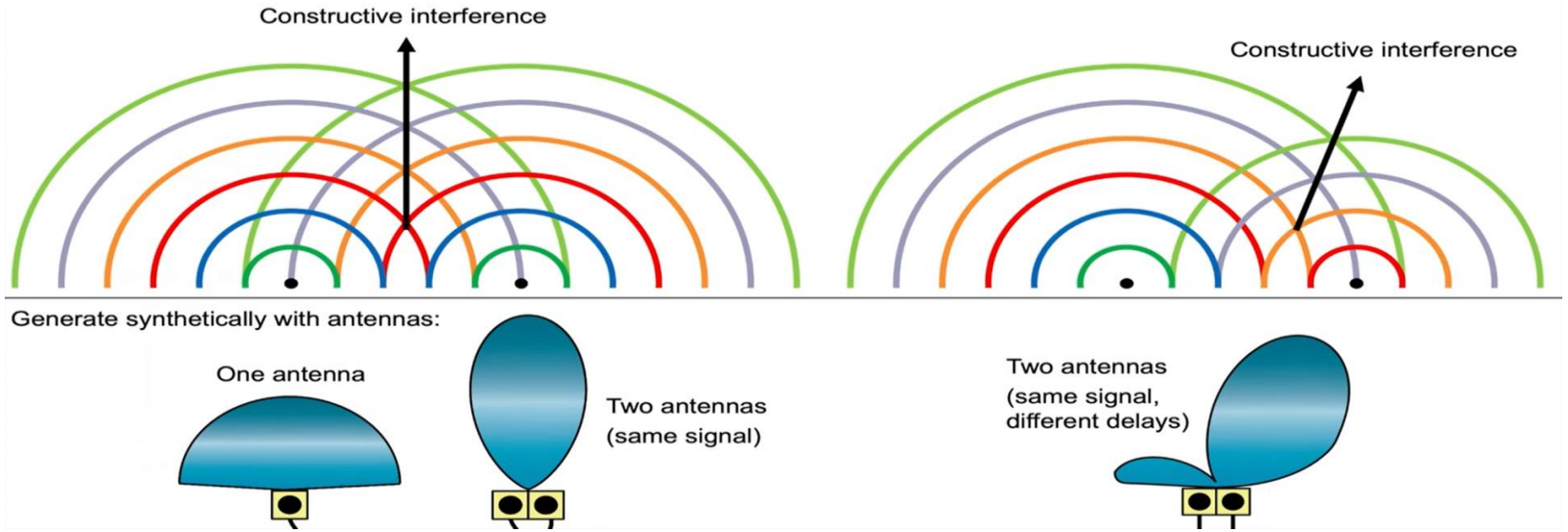
❑ Key enabling technology: Reconfigurable Intelligent Surface (RIS)

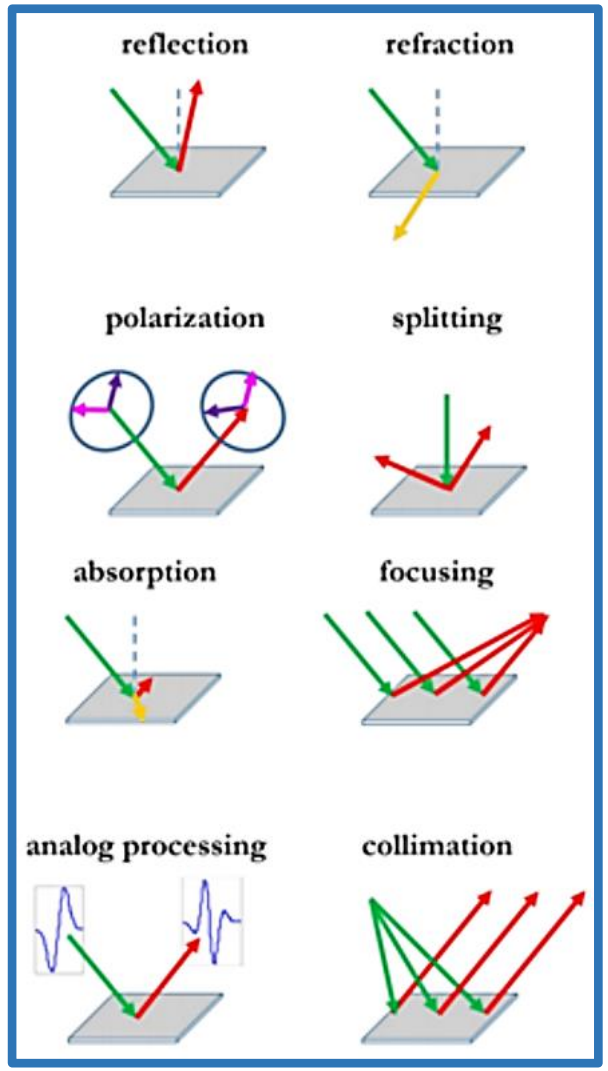
- Improve coverage and reduce fading
- Improve channel rank
- Deploy to get LOS channels

$$C = \log\left(1 + \frac{HP}{\sigma^2}\right)$$









Reference : [1] Shimin Gong, Xiao Lu , Dinh Thai Hoang, , Dusit Niyato, , Lei Shu, Dong In Kim, and Ying-Chang Liang , '' Toward Smart Wireless Communications via Intelligent Reflecting Surfaces: A Contemporary Survey '' , IEEE Communications Survays and Tutorials, 2020

RIS in 6G networks

Active RIS

Phase and amplitude modulation

Higher system gain with fewer elements; needs extra supply power

Passive RIS

Phase-only reflection, no extra power

Suffers from “multiplicative fading”: cascaded path loss limits capacity gain

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Background: RIS meets the UAV

Terrestrial RIS: limits

Serves only terminals on the same side of the surface

In cities, several RIS hops add path loss

Aerial RIS (ARIS)

Flexible deployment, high mobility

Creates line-of-sight links

360° reflection by UAV attitude + phase tuning

Open issues

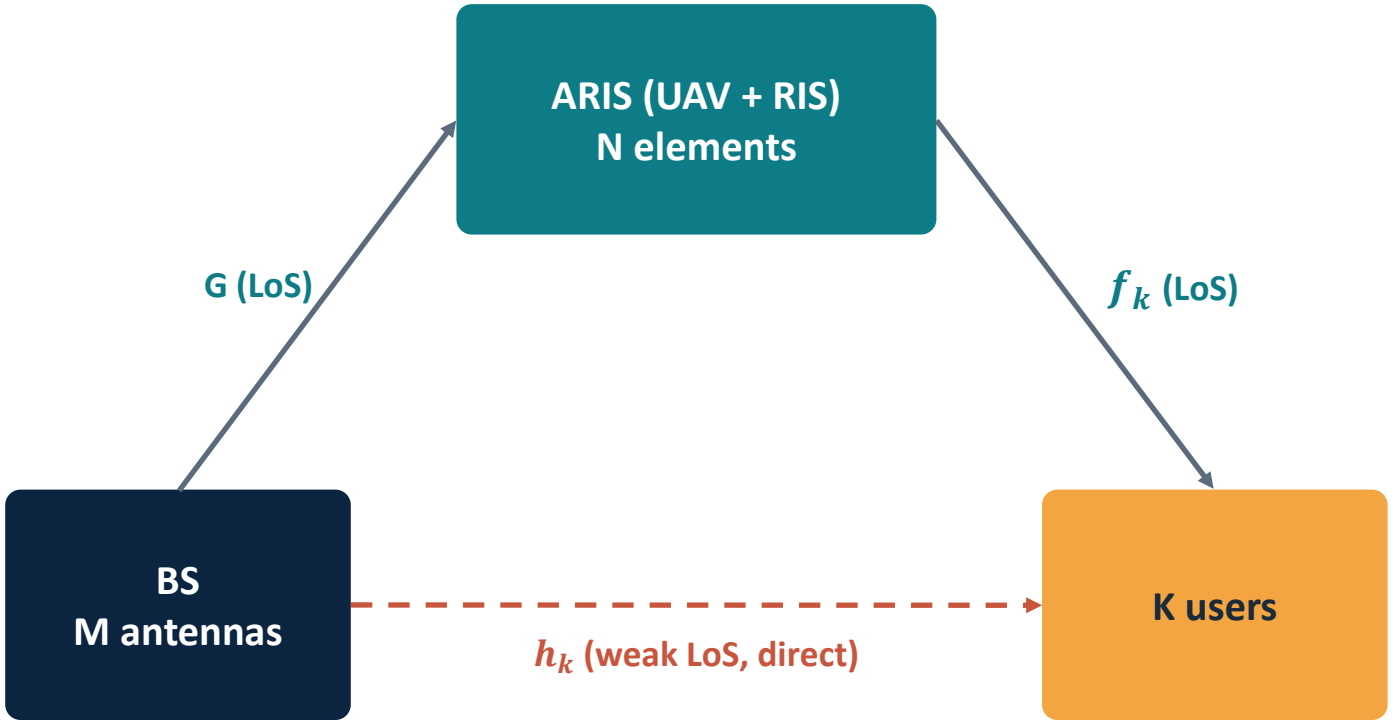
Buildings and trees weaken LoS to users

Active RIS needs extra power → UAV endurance

Paper's argument: amplification energy is much lower than the UAV's mechanical energy, so an active ARIS raises capacity without much cutting hover time.

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Basic problem: ARIS-aided system



Downlink MU-MISO, M > K

- Two paths per user: direct and indirect (via ARIS)
- ARIS has no RF chain → cannot estimate channels itself
- Cascaded channels estimated from pilots; the moving UAV makes CSI outdated
- Imperfect CSI: G′ = G+E, $f_k' = f_k + E_k$

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Basic problem: sum-rate maximization

$$\max_{w\phi} \quad \sum_{k=1}^{K} log_2(1+\gamma_k)$$

C1 BS power

$\|w\|^2 \leq P_{BS,max}$

C2 ARIS power

$\|\phi G w\|^2 + \|\phi\|\sigma_v^2 \leq P_{ARIS,max}$

C3 Phase shifts

$0 \leq \theta_n < 2\pi$ for all n

Non-convex, with w and ϕ coupled. Alternating optimization (fix ϕ → solve w, fix w → solve ϕ) needs many iterations as M, N, K grow, but the system must adapt in milliseconds.

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Definition and Development of SAC

What is Soft Actor-Critic?

SAC = an off-policy actor-critic deep RL algorithm built on maximum-entropy RL: the actor maximizes expected return and entropy, i.e. it succeeds at the task while acting as randomly as possible.

1

Actor-critic

Separate policy (actor) and value (critic) networks

2

Off-policy

Replay buffer reuses past experience → sample efficient

3

Entropy maximization

Encourages exploration and stability

Version used for continuous action spaces (e.g. beamforming weights)

References: [3] T. Haarnoja et al., Soft Actor-Critic: Off-policy maximum entropy deep RL with a stochastic actor, ICML 2018 (PMLR 80).
[4] T. Haarnoja et al., Soft Actor-Critic Algorithms and Applications, arXiv:1812.05905.
[5] OpenAI Spinning Up, Soft Actor-Critic documentation.

Why is SAC useful?

≈ 2 h

real-world quadruped walking learned from 160k steps

3 h vs 7.4 h

dexterous-hand valve task: SAC vs PPO

5 seeds

similar results across random seeds

- Sample efficient: off-policy learning from a replay buffer
- Stable: stochastic policy + entropy term; far less seed variance than a deterministic variant
- Little tuning: automatic temperature adjustment
- Matches or beats DDPG, PPO, TD3 and soft Q-learning, most clearly on the 21-D Humanoid

References: [3] T. Haarnoja et al., Soft Actor-Critic: Off-policy maximum entropy deep RL with a stochastic actor, ICML 2018 (PMLR 80).
[4] T. Haarnoja et al., Soft Actor-Critic Algorithms and Applications, arXiv:1812.05905.

How SAC emerged



TRPO / PPO
On-policy: stable, but fresh samples for almost every update

DDPG
Off-policy, deterministic actor; brittle to hyperparameters

Soft Q-learning
Max-entropy in deep RL, but sampler-based; not true actor-critic

SAC (ICML)
Off-policy max-entropy actor-critic + convergence proof

SAC v2 (Alg. & Apps)
Twin Q, no V net, auto temperature, real robots

References: [3] T. Haarnoja et al., Soft Actor-Critic: Off-policy maximum entropy deep RL with a stochastic actor, ICML 2018 (PMLR 80).
[4] T. Haarnoja et al., Soft Actor-Critic Algorithms and Applications, arXiv:1812.05905.
[5] OpenAI Spinning Up, Soft Actor-Critic documentation.

Definition: the maximum-entropy objective

$$\pi^* = \operatorname{argmax} \sum_t \mathrm{E}[\, \mathrm{r}(s_t, a_t) + \alpha \cdot \mathrm{H}(\pi(\cdot|s_t)) \,]$$

$$\mathrm{V}(\mathrm{s}) = E_{a \sim \pi}[\, \mathrm{Q}(\mathrm{s},\mathrm{a}) - \alpha \log \pi(\mathrm{a}|\mathrm{s}) \,]$$

$$T^{\pi}\mathrm{Q}(\mathrm{s},\mathrm{a}) = \mathrm{r}(\mathrm{s},\mathrm{a}) + \gamma\, E_{s'}[\, \mathrm{V}(\mathrm{s}') \,]$$

Why it helps

- Explores widely, gives up on clearly bad options
- Keeps several near-optimal behaviours (multi-modal)
- Robust to model and estimation errors
- Natural fit for uncertain CSI

α = temperature; α → 0 recovers standard RL. H = entropy of the policy.

References: [3] T. Haarnoja et al., Soft Actor-Critic: Off-policy maximum entropy deep RL with a stochastic actor, ICML 2018 (PMLR 80).
[4] T. Haarnoja et al., Soft Actor-Critic Algorithms and Applications, arXiv:1812.05905.
[5] OpenAI Spinning Up, Soft Actor-Critic documentation.

From soft policy iteration to SAC

Soft policy evaluation

Apply the soft Bellman operator repeatedly

Q_k → soft Q-function of π (Lemma1)

→

Soft policy improvement

π_{new}= argmin KL(π′ ‖ exp($\frac{Q}{a}$)/Z)

$Q^{new} \geq Q^{old}$(Lemma 2)

→

Convergence

Alternating both steps converges to the optimal max-entropy policy in the class Π (Theorem 1)

Practical SAC: replace tabular steps by neural networks and stochastic gradient descent, alternating one gradient step on the critics and one on the actor.

References: [3] T. Haarnoja et al., Soft Actor-Critic: Off-policy maximum entropy deep RL with a stochastic actor, ICML 2018 (PMLR 80).
[4] T. Haarnoja et al., Soft Actor-Critic Algorithms and Applications, arXiv:1812.05905.

Practical SAC: networks and updates

1 · Twin soft Q-critics

Minimize soft Bellman residual with target networks (τ = 0.005); use the min of the two Q values to curb overestimation

2 · Reparameterized actor

a = tanh(f_ϕ(ε; s)), Gaussian policy; loss J_π= E[α log π − Q]

3 · Automatic temperature

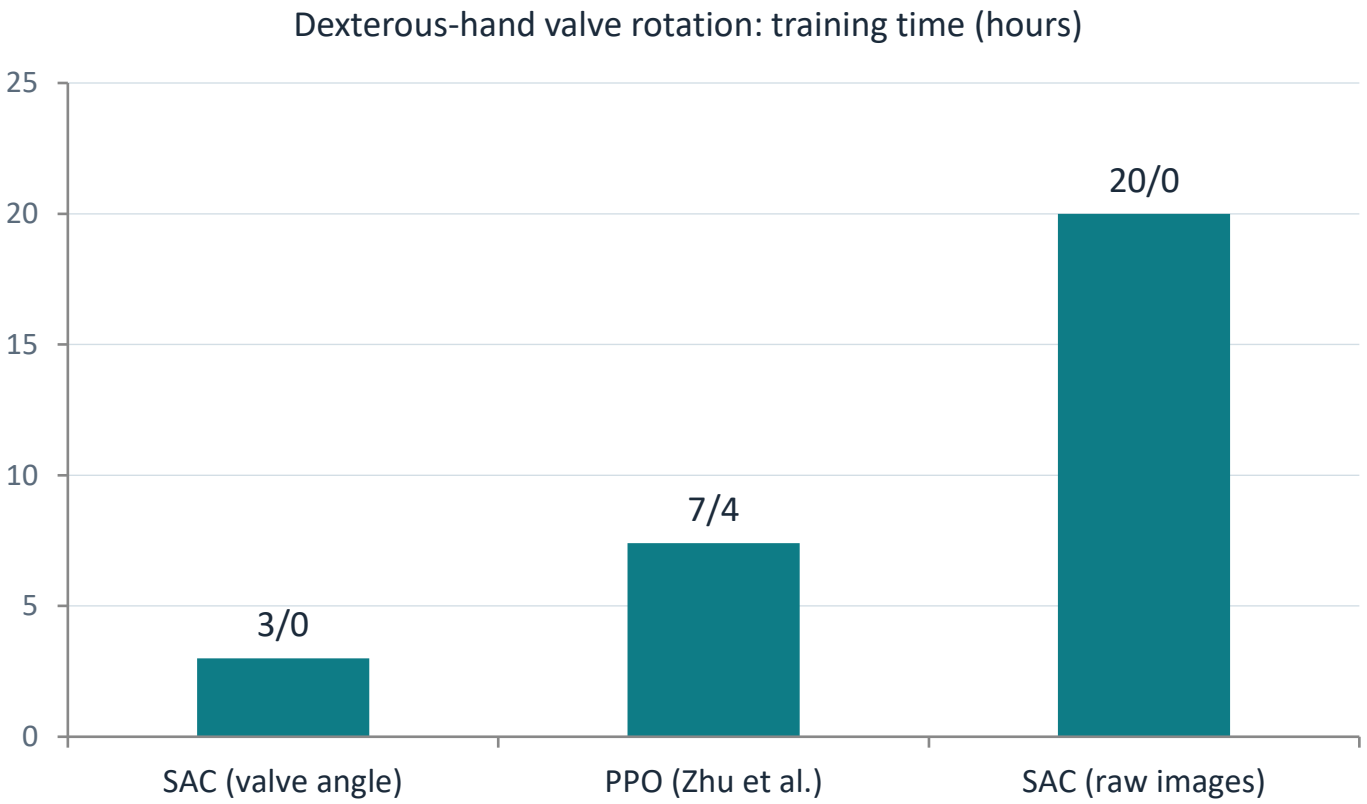
Learn α by minimizing J(α) = E[−α log π − α·Ħ], with target entropy Ħ = −dim(A)

4 · Replay buffer

Store (s, a, r, s′); sample mini-batches; one env step + one gradient step. Typical: lr $3 \cdot 10^{-4}$, γ = 0.99, batch 256, 2×256 hidden

References: [4] T. Haarnoja et al., Soft Actor-Critic Algorithms and Applications, arXiv:1812.05905.
[5] OpenAI Spinning Up, Soft Actor-Critic documentation.

Evidence: SAC in the real world



160k steps

Minitaur walks after ≈ 2 h real-world training

300k steps

valve from raw RGB images, no simulation

Trained on flat ground only, the gait generalized to slopes, blocks and stairs.

References: [4] T. Haarnoja et al., Soft Actor-Critic Algorithms and Applications, arXiv:1812.05905.

Challenges

Challenges: deep RL side

1

Sample complexity

Model-free deep RL can need millions of steps; on-policy methods need fresh data

2

Brittleness

Learning rates and exploration constants need per-task tuning (e.g. DDPG)

3

Stability

Off-policy + neural networks + continuous actions make convergence hard

4

Temperature / reward scale

Too small → near-uniform policy; too large → near-deterministic, poor exploration

SAC's answers: replay buffer, entropy-regularized stochastic actor, twin critics, automatic α.

Challenges: RIS-UAV side and our answer

Challenge	Answer in the article
Multiplicative fading limits passive RIS	Active ARIS amplifies the reflected signal
Imperfect CSI from UAV mobility	Max-entropy stochastic policy, trained with noisy G' and f'
Non-convex, coupled W and Φ; AO cost grows	Trained SAC policy outputs {W, Φ} directly
UAV endurance vs extra power	Few elements (N = 30), 1 % of power at the ARIS

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Article

A soft actor–critic reinforcement learning approach for over the air active beamforming with reconfigurable intelligent surface

The article at a glance

1 **Model**

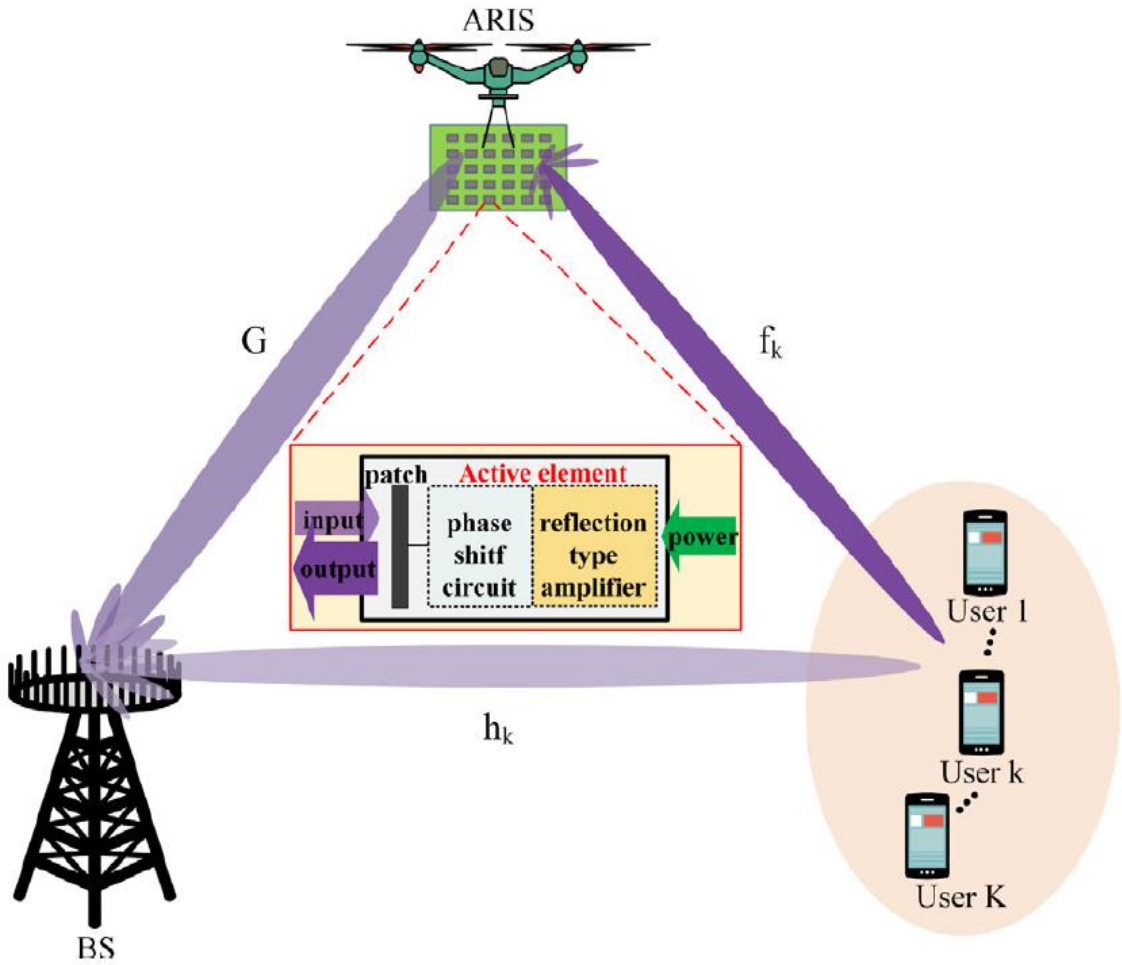
Active ARIS, weak LoS, imperfect CSI; maximize sum rate under power and phase constraints

2 **Method**

Recast as an MDP; SAC learns the BS precoder **W** and ARIS reflection $\boldsymbol{\phi}$ jointly

3 **Results**

Similar sum rate with perfect / imperfect CSI; ≈ 71–75 % above passive RIS



Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

system model

$$x = W s$$

$$y_{ARIS} = \Phi G x + \Phi v + n, \quad \Phi = \mathrm{diag}(\beta_1 e^{j\theta_1}, \ldots, \beta_N e^{j\theta_N})$$

$$y_k = (h_k^H + f_k^H \Phi G) W s + f_k^H \Phi v + n_k$$

$$y_k = \|H w_k\|^2 / (\Sigma_{j \neq k} \|H w_j\|^2 + \| f_k^H \Phi \|^2 \sigma_v^2 + (\| f_k \|^2 + 2)\sigma^2),$$
$$H = h_k^H + f_k^H \Phi G$$

- $\beta_j \in [0,1]$ amplitude, $\theta_i \in [0,2\pi)$ phase
- v: dynamic noise from amplifiers; n: static noise
- Direct link h_k mixes LoS and NLoS (factor κ)
- Imperfect CSI: G′ = G + E, $f_k' = f_k + E_k$, entries CSCG(0, σ_e^2)

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

SAC formulation as an MDP

State s_t
Powers (BS, ARIS, users), previous **W**, **Φ**, channels G, f_k (or G′, f_k'). D_k= 2(K + NM + NK + MK + N)

Action a_t
W and **Φ**, split into real and imaginary parts; dimension 2(MK + N)

Reward r_t
Sum rate Σ $\log_2(1 + \gamma_k)$ from channels, **W** and **Φ**

Replay buffer
Tuples (s_t, a_t , r_{t+1}, s_{t+1}) for off-policy updates

Neural networks cannot take complex values, so all complex matrices are split into real and imaginary parts.

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

The SAC algorithm

Environment: channels, BS, ARIS → Sample $a_t \sim \pi(\cdot|s_t)$, observe r_t , s_{t+1} → Store in replay buffer M → Mini-batch B → targets

Policy network π_ϕ State → mean and log-std (clipped to [−20, 2]) → Gaussian action

Q network (two heads) Soft Bellman residual J_Q ; KL-based policy loss J_π via reparameterization

Value network V (+ target) Squared residual J_v ; target updated softly with coefficient τ

Every few episodes the policy is evaluated. Output: optimal action a = {**W, ϕ**}.

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Simulation setup

Geometry

BS at (0, −60, 0) m; user centre at (300, 0, 0) m

K = 4 users within 50 m (also 5 m); ARIS hovers near (300, 10, 10) m

Channel and power

Path-loss exponent 3.6; weak LoS: 41.2 + 28.7 log d

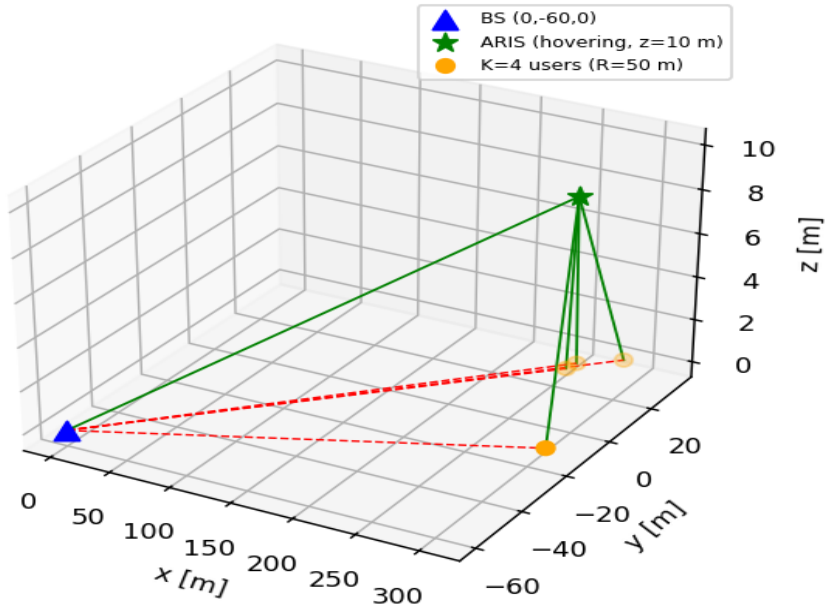
Noise −100 dBm; P_m= 30 dBm: 99 % BS, 1 % ARIS

Training

γ = 1, lr = 0.001, τ = 0.001, batch 16

Buffer 40,000; 40,000 steps; log-std clip (−20, 2)

Fig. 3 Simulation scenario (green = strong, red dashed = weak)



Five schemes compared

SAC perfect CSI · SAC imperfect CSI

Active RIS + AO · Passive RIS · No RIS (WMMSE)

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Results 1: training and convergence

10k vs 15k

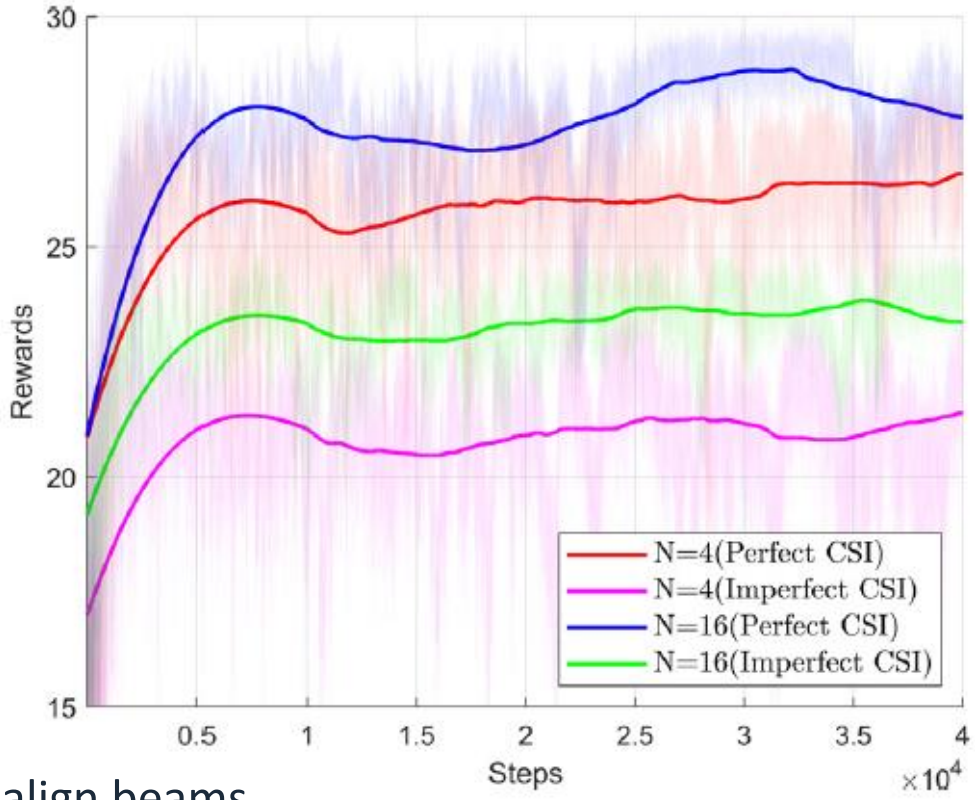
iterations to converge: lr 0.001 vs 0.0001

+25

reward gain from 30 to 50 dBm (N = 30)

27–29 vs 25–27

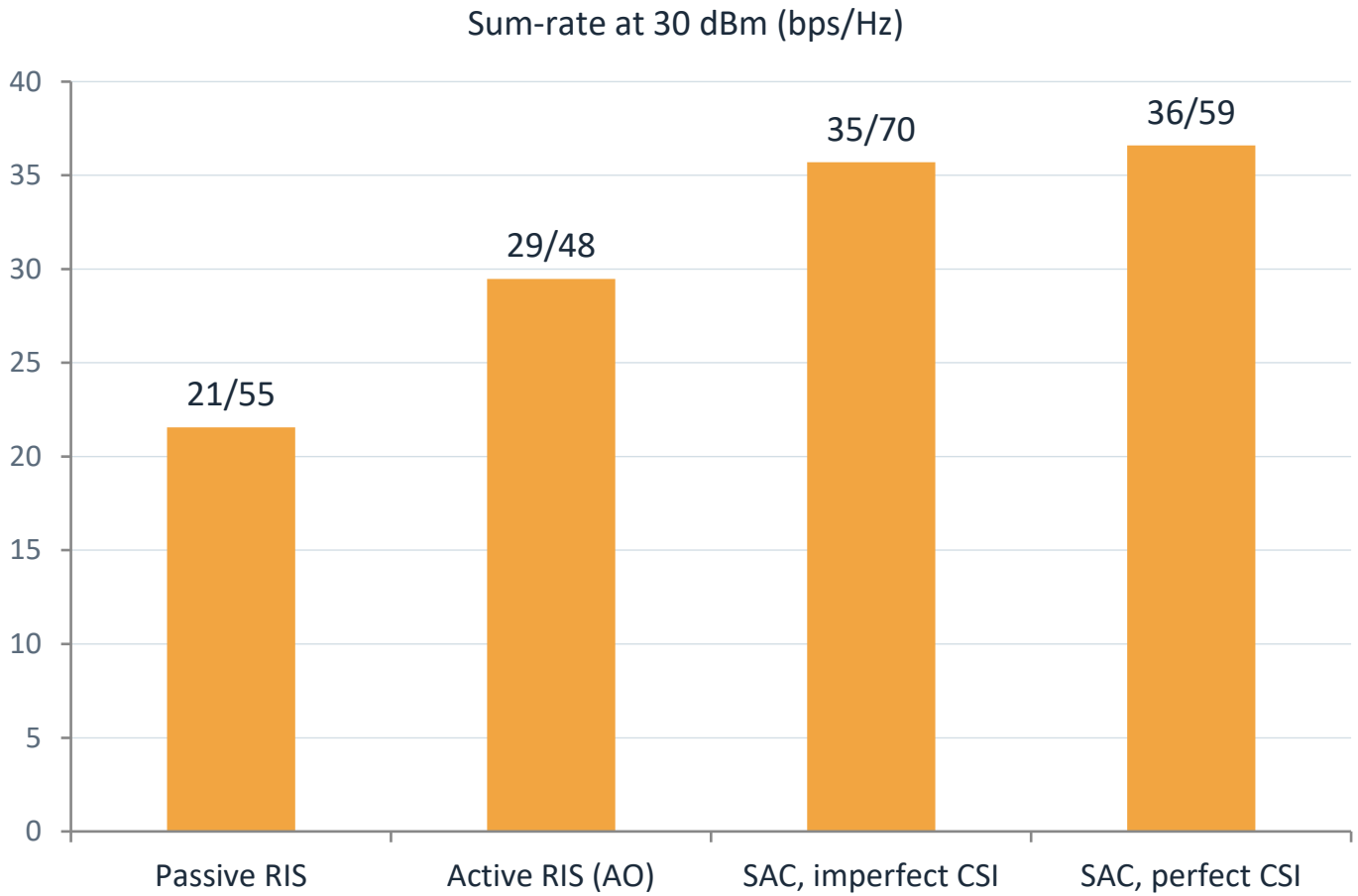
reward range, N = 16 vs N = 4 (perfect CSI)



- Perfect CSI always above imperfect CSI: outdated channels misalign beams
- lr = 0.001 adopted for all later experiments
- More reflecting elements and more power → higher sum rate (N from 10 to 130)

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Results 2: sum-rate versus baselines



+37 / +66 / +70 %

over passive RIS: AO / SAC imperfect / SAC perfect

N = 70 elements

Passive 21.95 · AO 32.85 · SAC 37.69 (imperfect) / 38.36 (perfect) bps/Hz

SAC gains grow faster with power: same rate with less power

Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Fig. 6 Sum-rate vs number of ARIS elements

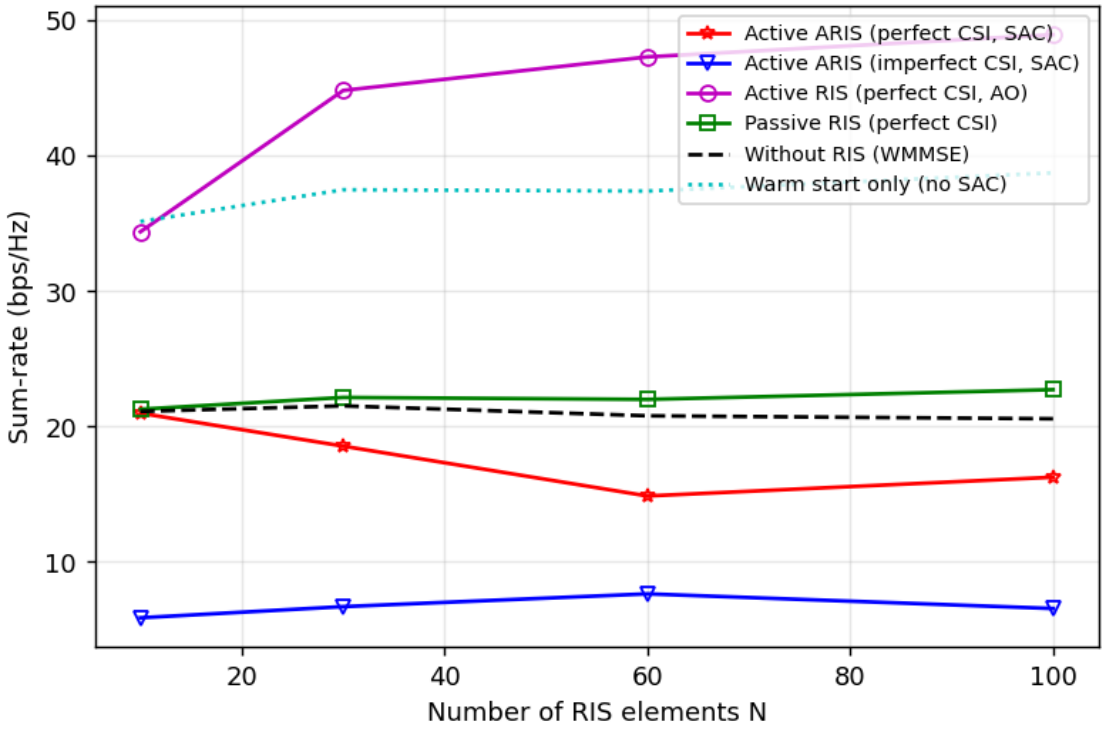
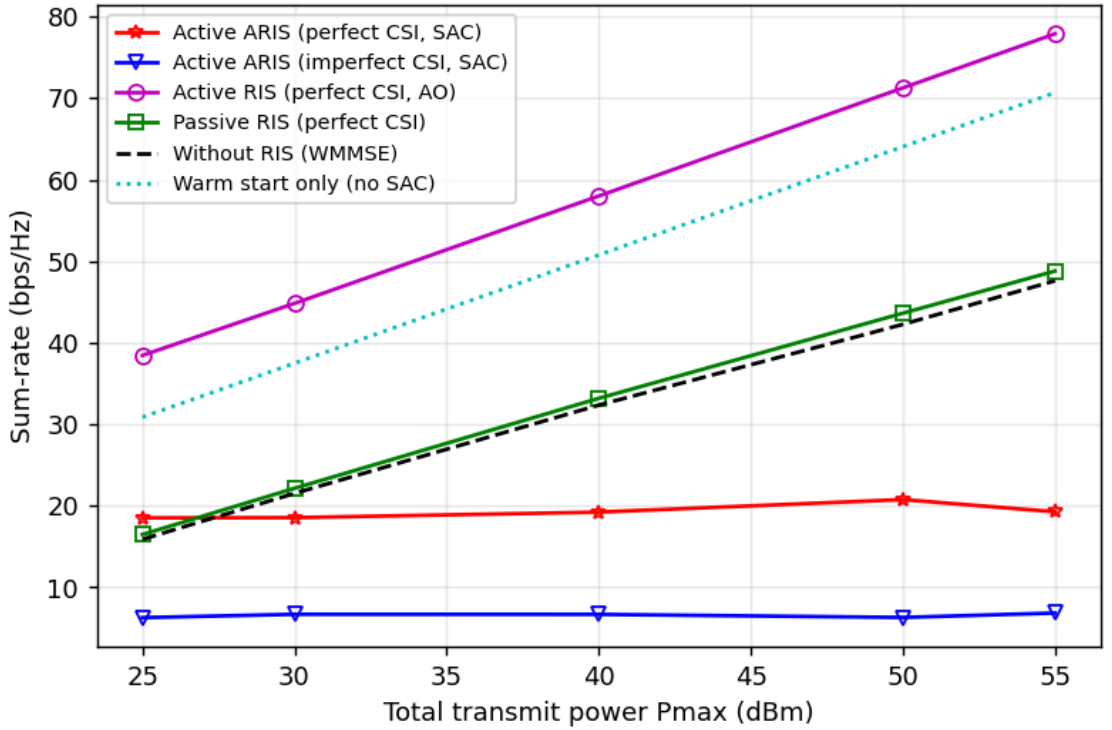


Fig. 8 Sum-rate vs total transmit power



Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Results 3: robustness to CSI and UAV position

Gap ≤ 0.89 bps/Hz

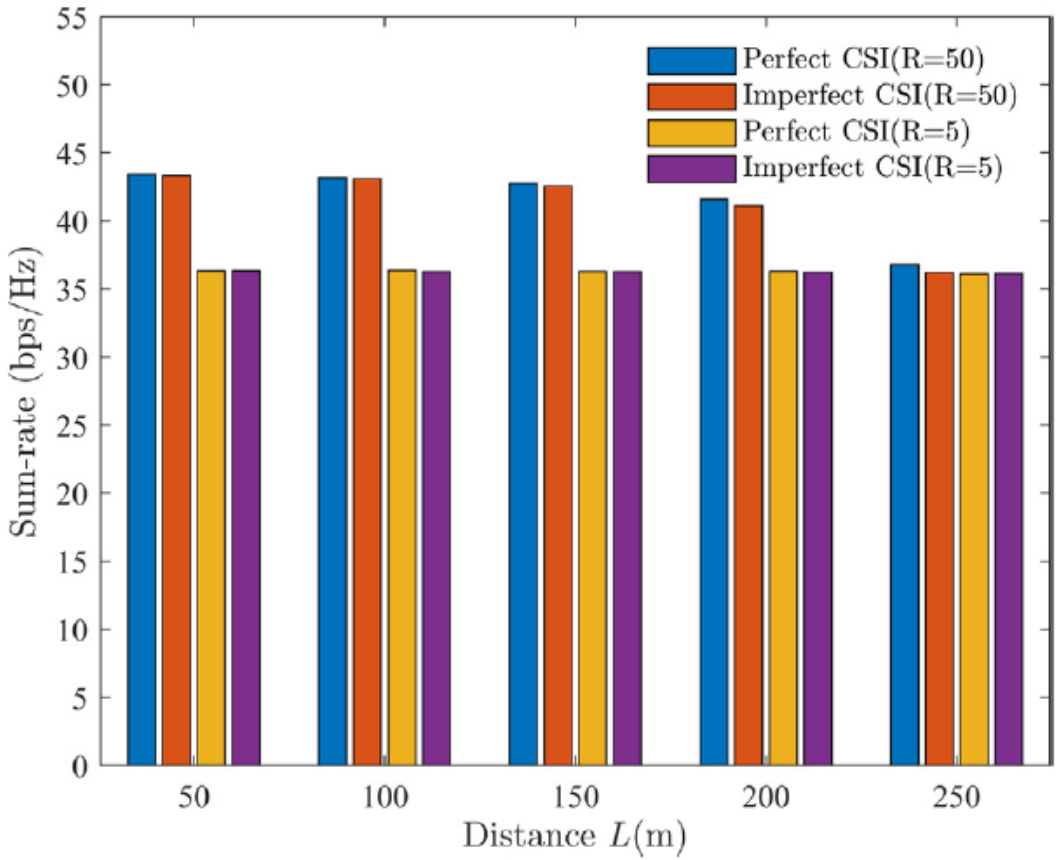
Perfect vs imperfect CSI, across all experiments

35.70 vs 29.48

SAC with imperfect CSI beats AO with perfect CSI (30 dBm)

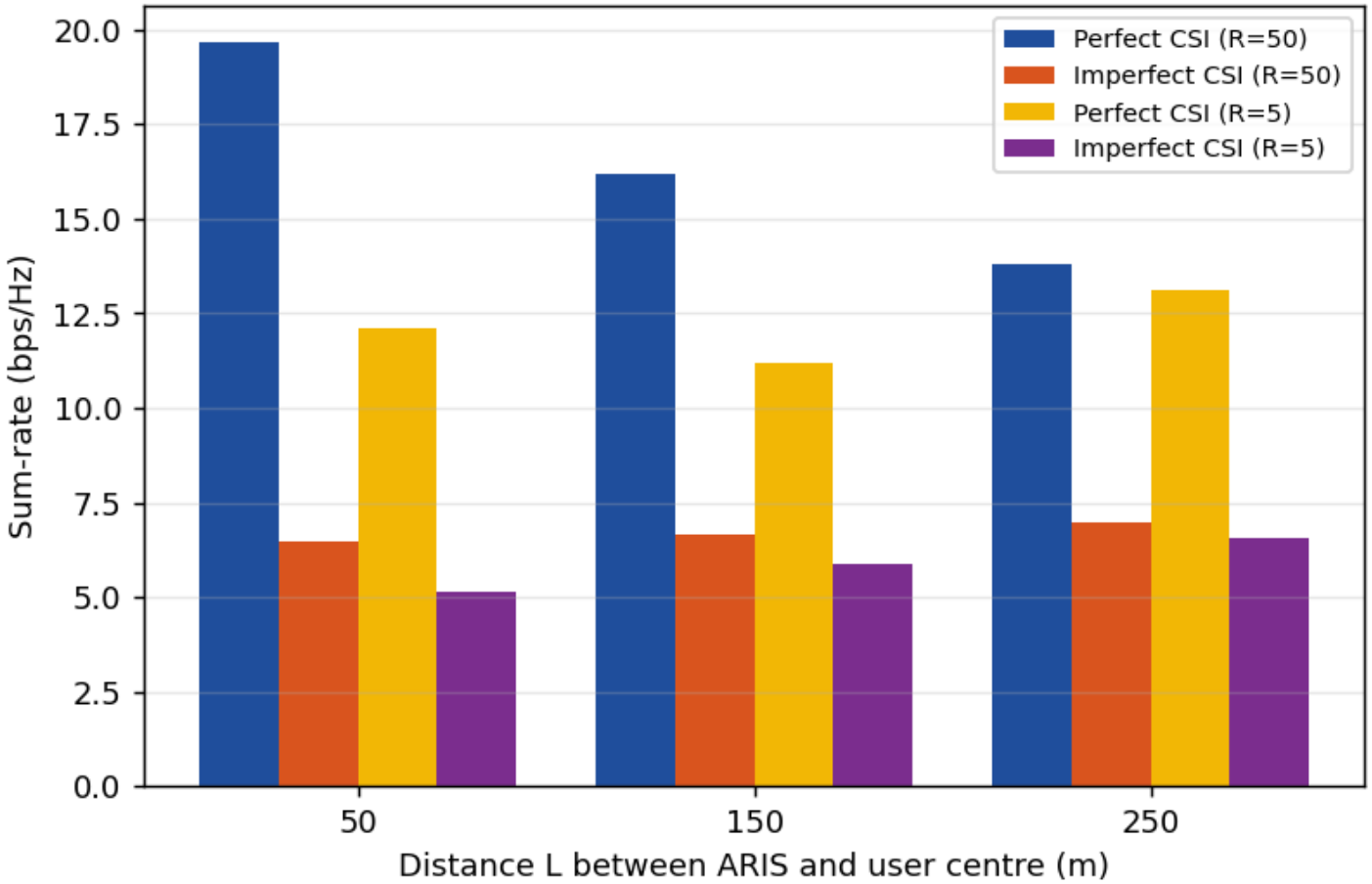
R = 50 m > R = 5 m

Users closer to the BS give higher rates; the perfect and imperfect curves nearly overlap



Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Fig. 9 Sum-rate vs ARIS position



Reference : [2] Z. Huang et al., A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS, Physical Communication 66 (2024)

Future work and Conclusion

Conclusion

- Active ARIS overcomes multiplicative fading
- SAC's max-entropy policy handles rapidly changing, imperfect CSI
- Jointly designs BS precoder W and ARIS reflection Φ
- Beats AO-based active RIS, same rate with less power
- Stable as the UAV position changes

Future Directions

- Long-endurance UAV use in active RIS-assisted communication
- Compare with TD3 / PPO (paper uses only AO baselines)
- Use SAC v2: twin critics, automatic α
- Add amplifier energy to the reward; optimize trajectory

[1] Shimin Gong et al., '' Toward Smart Wireless Communications via Intelligent Reflecting Surfaces: A Contemporary Survey '' , IEEE Communications Survays and Tutorials, 2020

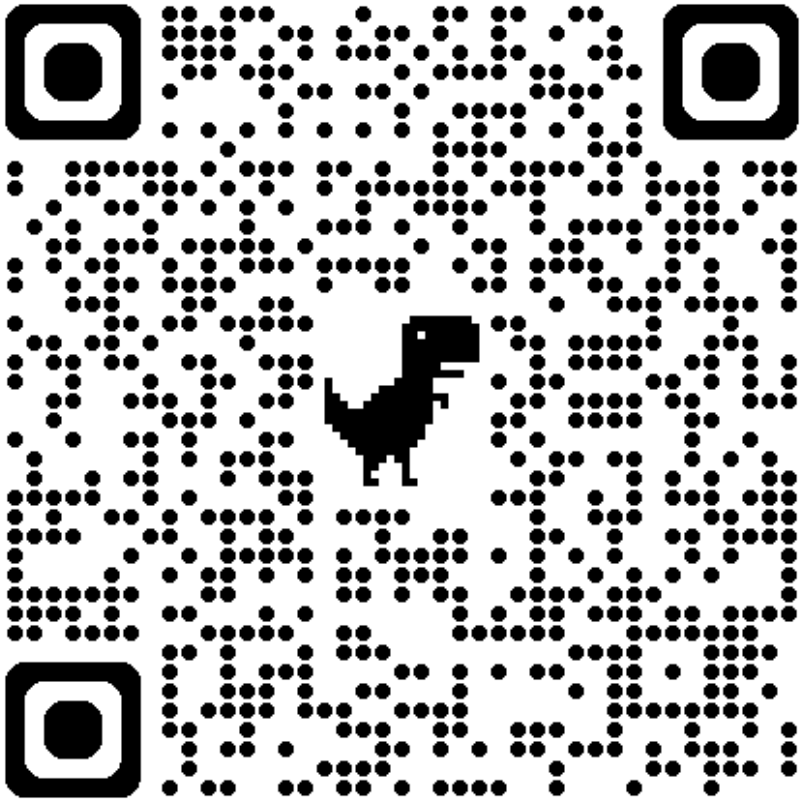
[2] Z. Huang et al., ‘’A soft actor–critic reinforcement learning approach for over the air active beamforming with RIS’’, Physical Communication 66 (2024) 102474.

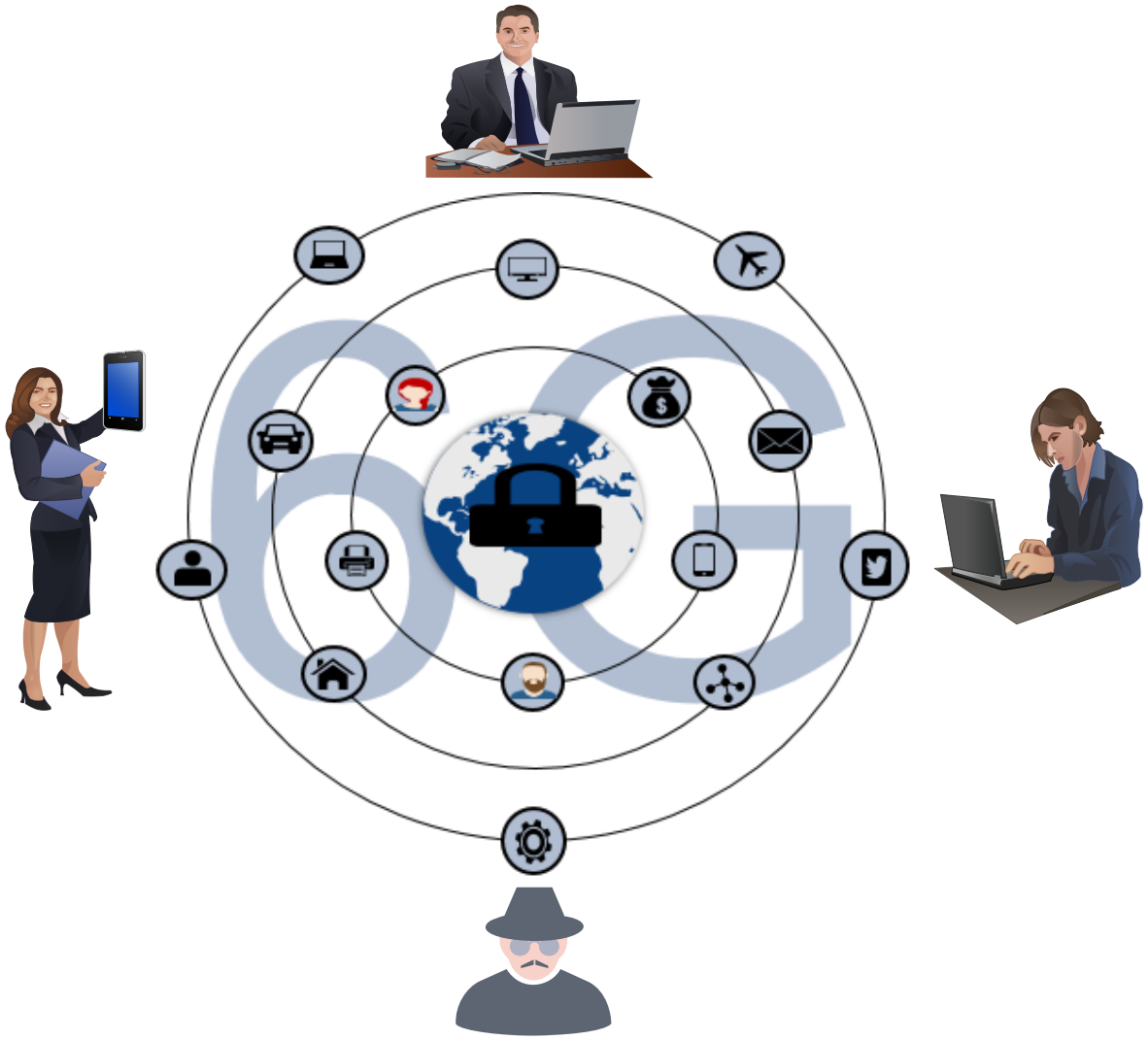
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[4] T. Haarnoja et al., ‘’Soft Actor-Critic Algorithms and Applications’’, arXiv:1812.05905.

[5] OpenAI Spinning Up, Soft Actor-Critic documentation.

https://forms.gle/L4363qJMHKteabUR6





Thank you for your attention